

Supplementary Material - 360° RAYS

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OVERVIEW

This supplementary document provides additional material for 360° RAYS. It starts with a comparison to related visualisation techniques, highlighting both similarities and the key differences that distinguish RAYS. It then describes the development of the method, including angular mapping, point placement, and path construction, along with some modified versions and features of RAYS.

1 RELATED WORK: FAMILIARITY AND DIFFERENCES COMPARED TO RAYS

1.1 Parallel Coordinates [5]

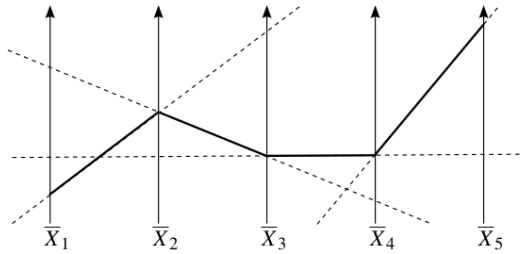


Figure 3: Constructing parallel coordinates with five dimensions represented by $N = 5$ vertical lines. Points in the plane are represented by lines joining the corresponding coordinates at the respective axes. Typically, only the line segments between the axes are drawn (represented by the bold poly-line).

Figure 1: Parallel Coordinates plot [5].

Each dimension is a vertical axis, and each data point is a poly-line connecting all axes. Both RAYS and Parallel coordinates (Figure 1) represent every dimension and every data point as a polyline.

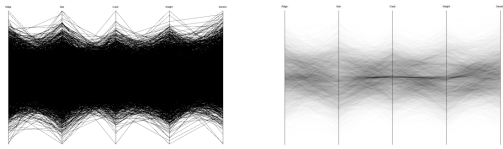


Figure 11: A frequently used synthetic dataset for parallel-coordinates plots about the geometric features of pollen grains. The large number of lines (3848 observations, left) may hinder the perception of patterns in the data. Note, however, that the hyperbolic shape of the envelope hints at normally distributed data. Density-based approaches (right) may reveal patterns that are not visible otherwise. The cluster of samples that appears in the center forms the word "EUREKA" if viewed in a scatterplot.

Figure 2: Parallel Coordinates plot [5], illustrating the high susceptibility of PC to visual clutter as data volume increases.

Parallel coordinates (PCP) place the axis linearly, producing extreme edge criss-crossing, and are susceptible to visual clutter

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as data volume increases (Figure 2). RAYS constrains each path within its angular sector, reducing the criss-crossing (**property t_6**). PC has no bitwise structure; RAYS encodes sign patterns via bitwise sector labels to make point identification easier (**property t_2**). One major difference between PCP and RAYS is that PCP encounters screen size limitations due to linear growth of the plot in one direction only, whereas RAYS solves this issue by using a radial structure.

1.2 Circle Segment [1]

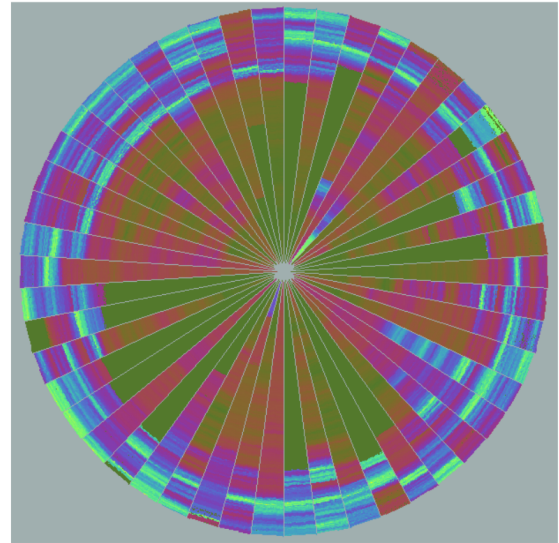


Figure 5: Representing about 265,000 50-dimensional Data Items with the 'Circle Segments' Technique

Figure 3: Circle Segments plot [1].

A pixel-per-value visualisation technique (Figure 3) represents each dimension as a segment of a circle and maps every data value to a colored pixel, so the geometric structure (orthant, sign pattern) is hidden inside coloured pixels. RAYS exposes this structure (**property t_1**). Circle Segment cannot show empty orthants, whereas RAYS can remove empty sectors to reveal sparsity immediately (**property t_1**). There is no path continuity per point, while RAYS guarantees a unique path from r_1 to r_d (**property t_2**).

Circle Segments requires at least 3 dimensions, and interpretation is dependent on how users assign dimensions to segments.

1.3 Concentric Circles [3]

Concentric Circle Coordinates is a generalisation of Parallel Coordinates designed to reduce visual clutter and utilise screen space efficiently.

Both Concentric Circles (Figure 4) and 360° RAYS are radial generalisations of Parallel Coordinates using concentric rings as dimension axes. Both use polycurves to represent individual data points. Both aim to reduce overlap compared to PC via the radial layout.

Concentric Circles place all points on the same full circle per ring, so paths still heavily overlap. RAYS *hierarchically partitions*

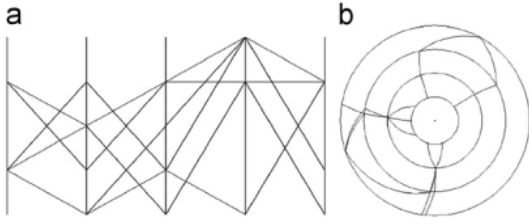
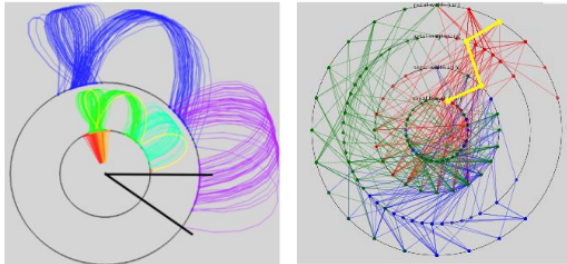


Fig. 1. The layout of parallel coordinates and concentric-circle coordinates: (a) Parallel Coordinates and (b) Concentric-circle Coordinates.

Figure 4: Concentric Coordinates plot [3].

each ring into angular sectors; each point is confined to a narrowing angular region, eliminating inter-sector crossings (**property** t_6). No sector-level purity or sparsity removal (**property** $t_1 - t_4$). Closed contours increase occlusion; RAYS open-arc sectors preserve point identity (**property** t_2). Another benefit of RAYS over Concentric Circles is that RAYS can show order-conditioned orthant utilization, highlighting the importance of individual features (**property** t_7). However, Concentric Circles have an edge over RAYS in terms of the space allocated to each feature in the radial structure, as with the increase in dimensions, the number of sectors increases exponentially, leaving a very minute angle for the place in the outer rings to be placed. By removing empty sectors and refactoring the sector angles while maintaining the hierarchical nature of the rings and sectors, RAYS can mitigate this issue, but while comparing static forms of both the visualisations, scalability remains a challenge.

1.4 Data Classification in Concentric Coordinates [13]



(a) Dynamic Circular Coordinates. Outer circle shows blue class, inner circle shows red class (outside) and green class (inside). One green class case is highlighted yellow.
(b) Concentric Coordinates so each circle is one attribute, each class is shown by class-colored polylines over all circles, one red class case is highlighted yellow.
Fig. 2. Three classes of 4-D Fisher Iris data visualized in Circular Coordinates (a) and Concentric Coordinates (b).

Figure 5: Data Classification in Concentric Coordinates plot [13].

Both Data Classification in Concentric Coordinates (Figure 5) and 360° RAYS are lossless high-dimensional visualisations using a concentric circles plotting method to reduce occlusion and improve classification and interpretability without dimensionality reduction. Both use concentric rings and colour-encode class labels.

Data Classification in Concentric Coordinates colours connecting *lines* by class. Along with using the same technique, RAYS additionally colours entire *sectors* as pure/mixed, enabling sector-level purity analysis (**property** t_3). RAYS uses CoV-based dimension ordering, but Data Classification in Concentric Coordinates notes ordering sensitivity as an open problem. RAYS maps directly to decision-tree pruning logic (**property** t_4), enabling understanding of subspace pruning and a map of classification logic. RAYS supports bar-graph-like density per sector (**property** t_6), enabling in-depth analysis of point distribution in orthants, which Data Classification in Concentric Coordinates lacks, as it only provides a

high-level analysis of the data.

1.5 RadViz and RadViz++ [10]

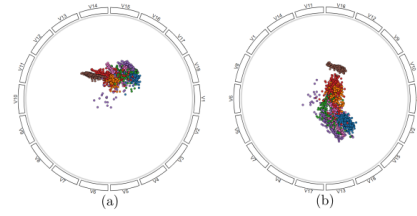


Figure 3. (a) RadViz with no variables ordering. (b) in RadViz++, anchors are rearranged in the circle according to their correlation coefficient. In our implementation, anchors are depicted by cells with the corresponding variable names above them, and points are colored based on their classes.

Figure 6: RadViz & RadViz++ [10].

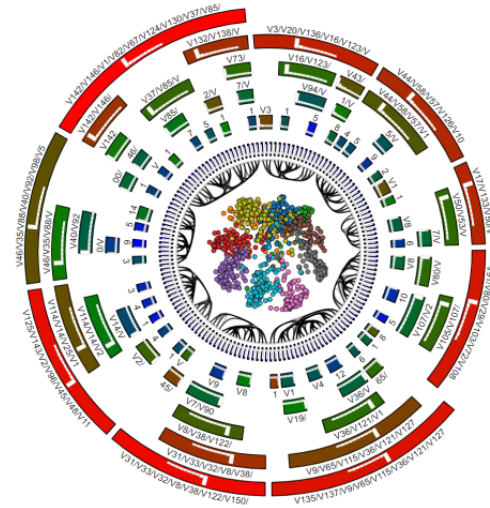


Figure 17. Core dataset visualized using RadViz++.

Figure 7: RadViz++ [10].

It is a radial visualisation used for data-to-variable and variable-to-variable analysis, which maps n variables as anchors along a circle, and data points are “pulled towards these anchors proportionally to their values.

Both visualisations - RadViz++ (Figure 7) and 360° RAYS aim to identify clusters and outliers relative to specific variables.

But RadViz is subject to visual ambiguities as RadViz collapses all dimensions into a single 2D point, erasing per-dimension structure; RAYS retains per-dimension values enabling orthant analysis. Also, high levels of aggregation can lead to overlap. Whereas RAYS is only dependent on the ordering of dimensions, which is at the user’s discretion, based on statistical methods like the coefficient of variation.

1.6 Comparison

RAYS distinguishes itself from existing multi-dimensional plotting techniques in many ways. RAYS plots all data points where the concentric rings represent dimensions ordered by coefficient of variation, conceptually acting as parallel coordinates bent into circles. The edges connecting successive points/dimensions split based on point values, resembling a dendrogram-like branching [9], which helps identify data trends; at the final ring, the sector usage and class distribution (Property t_1) can be immediately assessed,

similar to analyzing cluster formation in a dendrogram. Compared to Parallel Coordinates, the rings in RAYS act as the axes for the data, and the coefficient-of-variation-based dimension ordering significantly aids in clutter reduction and pattern stability, addressing a major challenge where traditional parallel coordinates often struggle due to insufficient space and overlapping lines [6]. Furthermore, unlike visualisations like star coordinates [8], which map multi-dimensional data to 2D for cluster identification, RAYS preserves the raw, original information for all d dimensions, preventing information loss and ensuring the visualisation reflects genuine patterns in the $\mathbb{R}^d$ space. RAYS provides a datapoint-level analysis by showing the precise path and position of every point x_i , contrasting with many pixel-based methods like circle segments that only offer a high-level aggregate view of the dataset.

2 RAYS DEVELOPMENT

2.1 Angular Mapping and Point Placement Calculation

The visualisation maps the point's coordinate value x_{ij} onto the specific angular range of the sector it belongs to. Let θ_{start} be the starting angle of the sector and $\Delta\theta_{ij}$ be its total angular width. The angular position Θ_{ij} is then calculated by applying the normalized scaling factor V_{norm} to the sector's total angular width ($\Delta\theta_{ij}$): $\Theta_{ij} = \theta_{start} + V_{norm} \cdot \Delta\theta_{ij}$

2.2 Point Representation:

A point's location, up to the $j-1$ dimension, $(x_{i1}, \dots, x_{i(j-1)})$, is represented by a path that traverses from the sector corresponding to x_{i1} , to x_{i2} , and so on, up to $x_{i(j-1)}$ across the $j-1$ dimensions (rings). Once the j^{th} dimension is added, the path from $x_{i(j-1)}$ to x_{ij} proceeds based on the value (non-negative or negative) of the j^{th} coordinate.

The final representation of the complete d -dimensional point $x_i = (x_{i1}, \dots, x_{id})$ is the entire path extending from the innermost concentric circle (dimension a_1) to the outermost concentric ring (dimension a_d). This path's angular extent uniquely identifies **orthant** to which the point x_i belongs in the $\mathbb{R}^d$ space.

2.3 Algorithm

Using the terminology defined previously, we apply the following algorithm to set up the **RAYS** visualisation structure:

1. **Input:** $\mathbf{X}$, a d -dimensional dataset with n points, $\mathbf{X} = \{x_1, x_2, \dots, x_n\}$.
2. **Point Format:** Each point x_i is represented as a vector: $(x_{i1}, x_{i2}, \dots, x_{id})$ where, $x_{ij} \in [-1, 1]$ for $1 \leq i \leq n$ and $1 \leq j \leq d$ (after normalisation).
3. **Dimension Ordering:** The indices $j = 1, \dots, d$ correspond to the dimensions $a_1, a_2, \dots, a_d$, which are ordered by non-decreasing coefficient of variation:

$$CV_{a_1} \leq CV_{a_2} \leq \dots \leq CV_{a_d}$$

Algorithm 1 RAYS Visualisation Construction

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1: procedure CONSTRUCTRAYS( $\mathbf{X}$ )
2:   Define the radius  $r_j$  and width  $W_j$  for each ring  $j = 1 \dots d$ .
3:   Ring Width Definition: The radius of ring  $j$  is  $r_j$  and its width is  $W_j = r_j - r_{j-1}$ .
4:   •  $r_0$  is defined as the center (0). And,  $W_j = W$  (constant width).
5:   Initialisation (Dimension  $a_1$ ):
6:   Create the innermost concentric circle (Ring 1) with radius  $r_1$ .
7:   Partition the circle into two sectors based on the sign of  $x_{i1}$ :
8:   • One sector  $S_{1,+}$  for  $x_{i1} \geq 0$  (e.g., the non-negative half).
9:   • One sector  $S_{1,-}$  for  $x_{i1} < 0$  (e.g., negative half).
10:  Place all points' first coordinates  $x_{i1}$  into the corresponding sectors of the innermost circle.
11:  Determine the initial angular position  $\Theta_{i1}$  for point  $i$  by scaling the value  $x_{i1}$  to the angular range of its determined sector.
12:  for  $j = 2$  to  $d$  do
13:    Iterative Partitioning (Dimension  $a_j$ ):
14:    Create the  $j^{th}$  concentric ring (radius  $r_j$ , width  $W_j = W$ ).
15:    for each existing path segment  $P_{i,j-1}$  ending at dimension  $a_{j-1}$  in ring  $j-1$  do
16:      Let  $S_{i,j-1}$  be the sector containing  $P_{i,j-1}$ .
17:      Subdivide the corresponding sector in ring  $j$  into two new sub-sectors  $S_{i,j,+}$  and  $S_{i,j,-}$ .
18:      The subdivision is based on the sign of the current coordinate  $x_{ij}$ :
19:      •  $S_{i,j,+}$  for  $x_{ij} \geq 0$ .
20:      •  $S_{i,j,-}$  for  $x_{ij} < 0$ .
21:      Point Placement and Path Extension:
22:      Calculate the angular position  $\Theta_{ij}$  for point  $i$  by scaling the value  $x_{ij}$  to the angular range of the determined sub-sector  $S_{i,j}$ .
23:      Place the point  $x_{ij}$  at the polar coordinates  $(r_j, \Theta_{ij})$ .
24:      Extend the path segment  $P_{i,j-1}$  to the new point position at  $(r_j, \Theta_{ij})$  in  $S_{i,j}$ .
25:    end for
26:  end for
27:  Output: A radial layout where each point  $x_i$  is represented by a unique continuous path from  $r_1$  to  $r_d$ , identifying its orthant/sector based on the bitwise sign encoding of its coordinates  $(x_{i1}, \dots, x_{id})$ . The points are located in its orthant/sector based on its value relative to its previous dimension's point position.
28: end procedure

```

2.4 Modified RAYS

2.4.1 Collapsed Empty Sectors

The handling of empty sectors in RAYS is done through two complementary approaches:

- **Removing Empty Sectors:**

- In this approach, sectors containing no data points are completely removed from the visualization.
- The removed sectors are replaced with space, making it easier to visually identify gaps in the distribution and quickly spot the sparsity of the points in the $\mathbb{R}^d$ space.

- **Collapsing Empty Sectors with Angle Refactoring:**

- This method not only removes empty sectors but also redistributes their angular space among the remaining occupied sectors.

- The reallocation is performed within each ring while preserving the hierarchical structure of the visualization.
- This is beneficial for higher-dimensional rings (outer rings), where the number of sectors grows exponentially, resulting in very small angular widths. By eliminating empty sectors and reallocating their angles, the visibility and interpretability of occupied sectors are significantly improved.

2.4.2 Compact View

The Compact View represents a pruned version of the RAYS visualization, inspired by the behavior of decision trees.

- Sectors are subdivided only if they are *impure* (i.e., contain data points from multiple classes).
- For a given ring R_j (corresponding to the j^{th} dimension in $\mathbb{R}^d$), if a sector is identified as either:
 - *pure* (containing points from a single class), or
 - *empty*,

it is not further split into child sectors in the subsequent ring R_{j+1} .

- This selective expansion reduces visual clutter and highlights only the regions of interest where class ambiguity exists.

2.4.3 Additional Features

To further improve usability and analytical clarity, the following features are incorporated:

- **Removing Connections:**
 - This feature disables the edge connections between sectors.
 - Removing these connections prevents visual occlusion, allowing the data points within each sector to be observed more clearly.
 - It is particularly useful for detailed, sector-wise inspection of point distributions (property t_5).
- **Label Filtering:**
 - This feature enables users to selectively display specific class labels in multi-class datasets.
 - Since multiple colors can lead to visual clutter, filtering helps isolate individual classes for better analysis.
 - This functionality has been validated through the user study as an effective way to improve interpretability.

3 USER STUDY

The study’s primary aim is the value-driven evaluation of 360° RAYS visualisation using the ICE-T heuristic framework [11, 12]. The study is guided by three research questions (RQ):

- **RQ1 (Insight):** To what extent does the visualization enable users to discover patterns, relationships, and unexpected data characteristics?
- **RQ2 (Time & Essence):** How efficiently does the visualization support quick browsing and targeted search, and how well does it communicate an overall understanding of each dataset?
- **RQ3 (Confidence):** How well does the visualization promote trust in the data representation and support learning about the broader domain context?

3.1 Participants and Setting

In our user study, we invited participants with knowledge of data visualisation and basic classification and clustering concepts. Those who lacked this background were excluded. The study included nine participants with engineering backgrounds, specialising in computer science, aged between 19 and 24.

We first briefed participants on the user study, its expectations, and the basic concepts required to understand 360° RAYS. After this introduction, participants were shown a sample visualisation constructed from the Iris dataset [2] to explain the visual elements on a 15.6-inch, 1920 × 1080, 60 Hz display. Before viewing each visualisation, participants also received a short description of the corresponding dataset. They then interacted with the visualisation using a mousepad. After reviewing all visualisations, each participant completed a standard survey form based on the ICE-T heuristic framework on a tablet using a stylus. The survey included 21 7-point Likert-scale questions covering ease of use, insight value, time required to assess the visualisation, the essence conveyed by the visualisation, and participant confidence. Each session lasted around 30- 45 minutes. Audio and video were recorded using Microsoft Teams recording and transcription tools. We collected data through direct observation, verbal reports, screenshot recordings, and participants’ self-assessed evaluations of confidence and simplicity for every task. A follow-up interview was then conducted to gather participants’ views on the effectiveness of the depiction and their overall level of satisfaction.

3.2 Dataset Details

The three datasets used for the study are:

- **Glass Identification (Forensic Science / Materials) [4]**
 - Records: 214
 - Variables: 10 (9 continuous, 1 categorical)
 - Key Characteristics: Refractive index and elemental composition (Na, Mg, Al, Si, K, Ca, Ba, Fe); classification into 6 glass types
- **Diabetes (Pima Indians) (Clinical Medicine / Epidemiology) [7]**
 - Records: 768
 - Variables: 9 (8 continuous, 1 binary)
 - Key Characteristics: Pregnancies; glucose; blood pressure; skin thickness; insulin; BMI; diabetes pedigree function; age; and a binary diabetic/non-diabetic outcome
- **Earthquake & Tsunami [14]**
Domain: Geoscience / Hazard Research
 - Records: 782
 - Variables: 13 (8 continuous, 5 ordinal/binary)
 - Key Characteristics: Magnitude (6.5–9.1), depth, location, CDI/MMI intensity, significance, year/month (2001–2022); binary tsunami outcome (304 events)

3.3 Hypotheses

Based on the ICE-T framework and the design of the novel visualization RAYS, the directional hypotheses proposed is the visualization will score at or above 5.0 (on the 7-point scale) on at least three of the four ICE-T components, indicating overall value.

3.4 Results

The Survey uses a 7-point Likert scale (1–7), where:

- 1 = Strongly Disagree
- 4 = Neutral
- 7 = Strongly Agree

The participant scores, averages of each component, and the total average are calculated in Table 1. We achieve a total average score of 5.712 (including all participants) and a score of ≥ 5 in each of the components of the survey. Based on the ICE-T standard, a visualization with a score of ≥ 5 is considered a good visualization.

3.5 Observation

The important features and resulting insights derived from the user study are as follows:

3.5.1 Interactive and Arrangement Features

- **Coefficient of Variation (CoV) Order:** Both ascending and descending CoV orders of rings were found to be important and provided key insights. Ascending CoV order helped users clearly prune features.
- **Interactive Features:** Features such as zooming, removing connection edges between points, and label filters significantly enhanced analysis.
 - Removing connection edges helped to analyze the positioning of data points in large datasets, along with providing a sector-wise analysis clearly.
 - Zooming helped in efficiently inferring data values and point locations in the orthants.
 - The label filter was beneficial in per-class analysis and reading data more clearly, especially in the dataset with a higher number of labels, for example, in the glass dataset.
- **Compact View:** This feature specifically helped in the case of ascending order of CoV, making it easier to prune features.
- **Positive-Negative Division:** This helped in quickly drawing inferences about the range of data values leading to a particular class.

3.5.2 Analytical and Interpretive Insights

- **Critical Feature/Subspace Identification:** Users were able to identify the critical subspace using the ascending CoV order and the locality of points. They could also identify critical features and important features for a specific class based on the density of edges/points as well as the purity of sectors(orthants).
- **Data Clustering and Classification:**
 - Classification of points was achievable (e.g., determining which features result in diabetes or not).
 - Sector-wise analysis was effective in interpreting the data and hidden inferences, like a lower concentration of elements forming a particular glass.
- **Outlier and trend Identification:**
 - Outliers, such as a glass of one type having a completely different composition, were identifiable visually.

- Edge and point density helped identify common trends in glass composition.

- **Understanding Feature Importance and Data Gaps & Bias:** Empty sectors helped users understand the importance of features, or indicated that data was either not recorded or a combination was not possible. Data sparsity could be understood using the bit vector classification and the division of space into subspaces.

3.5.3 Design Considerations/Limitations

- **Ring Importance:** Inner rings were perceived as more important and accurate for analysis, as the outer ring sectors' angle was negligible, making it difficult to read with the naked eye.
- **Visualization Scalability:** The visualization was noted as potentially detrimental for a large number of labels, although filtering can mitigate this limitation.

Survey Questions		Participant Scores									
Category	Question	1	2	3	4	5	6	7	8	9	Avg
Insight	The visualization exposes individual data cases and their attributes	6	7	6	4	6	6	6	6	6	
	The visualization facilitates perceiving relationships in the data, like patterns & distributions of the variables	7	7	4	7	6	7	7	5	5	
	The visualization promotes exploring relationships between individual data cases as well as different groupings of data cases	6	6	5	3	5	7	5	3	6	
	The visualization helps generate data-driven questions	6	7	6	4	7	7	4	5	7	
	The visualization helps identify unusual or unexpected, yet valid, data characteristics or values	5	7	6	5	6	7	7	5	6	
	The visualization provides useful interactive capabilities to help investigate the data in multiple ways	5	6	6	4	7	7	7	7	6	
	The visualization shows multiple perspectives about the data	5	7	7	6	6	6	7	6	6	
	The visualization uses an effective representation of the data that shows related and partially related data cases	6	6	4	6	7	7	6	4	5	
	$S_{insight}$ (Avg)		5.75	6.625	5.5	4.875	6.25	6.75	6.125	5.125	5.875
Time	The visualization provides a meaningful spatial organization of the data	5	6	3	6	6	7	6	6	6	
	The visualization shows key characteristics of the data at a glance	6	6	5	4	7	7	5	2	5	
	The interface supports using different attributes of the data to reorganize the visualization's appearance	7	7	6	6	5	7	7	5	7	
	The visualization supports smooth transitions between levels of detail	4	5	2	5	6	6	5	3	7	
	The visualization avoids complex commands and textual queries by providing direct interaction with the data representation	6	6	7	5	7	7	6	2	6	
	S_{time} (Avg)		5.6	6	4.6	5.2	6.2	6.8	5.8	3.6	6.2
Essence	The visualization provides a comprehensive and accessible overview of the data	6	6	6	5	7	7	4	5	5	
	The visualization presents the data by providing a meaningful visual schema	6	6	5	6	6	6	6	6	5	
	The visualization facilitates generalizations and extrapolations of patterns and conclusions	6	7	5	6	6	6	7	7	5	
	The visualization helps understand how variables relate in order to accomplish different analytic tasks	7	6	4	6	6	6	5	5	6	
	$S_{essence}$ (Avg)		6.25	6.25	5	5.75	6.25	6.25	5.5	5.75	5.25
Confidence	The visualization uses meaningful and accurate visual encodings to represent the data	7	5	4	5	7	7	6	5	6	
	The visualization avoids using misleading representations	7	7	6	5	6	7	4	6	7	
	The visualization promotes understanding data domain characteristics beyond the individual data cases and attributes	6	3	6	6	7	7	4	7	7	
	If there were data issues like unexpected, duplicate, missing, or invalid data, the visualization would highlight those issues	7	4	4	4	7	6	3	1	6	
	$S_{confidence}$ (Avg)		6.75	4.75	5	5	6.75	6.75	4.25	4.75	6.5
Total Avg Value										5.712	

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